Initial Evaluation of a Healthcare Dashboard

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1 Objective

The purpose of this report is to conduct an initial evaluation of a healthcare dashboard developed as part of a group project. The aim is to assess its structure, usability, and effectiveness in supporting data interpretation and decision-making.

2 Dashboard Overview

The dashboard presents a range of patient-related data, including demographic characteristics, symptom patterns, and activity levels. It incorporates multiple visual components such as:

- Key Performance Indicator (KPI) cards
- Donut charts for categorical distributions
- Scatter plots to explore variable relationships
- A waterfall chart for category changes
- A geographical map
- Profile-level breakdowns
- Interactive slicers and navigation elements

The dashboard is designed to enable exploratory analysis through a variety of visual and interactive tools.

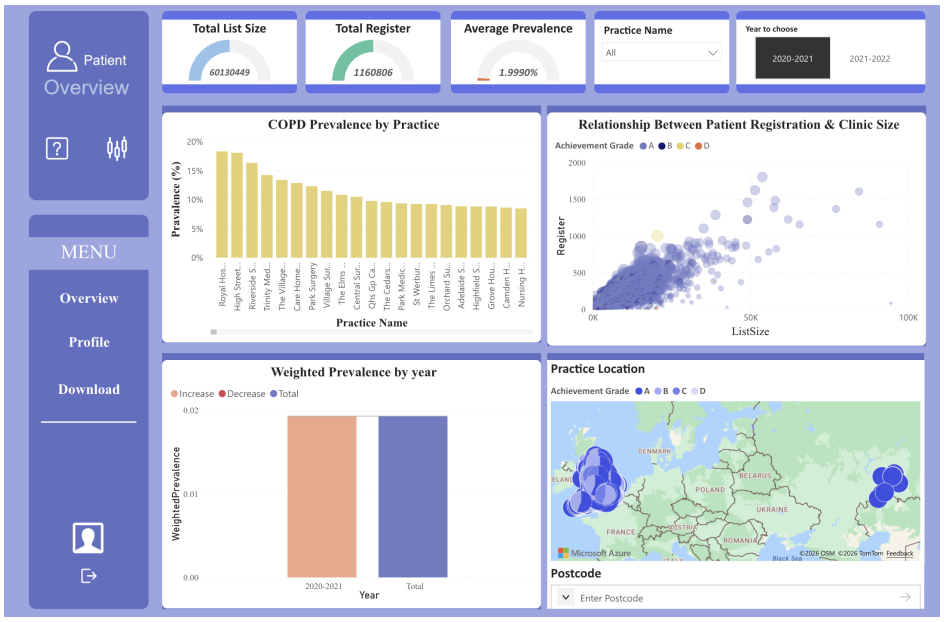
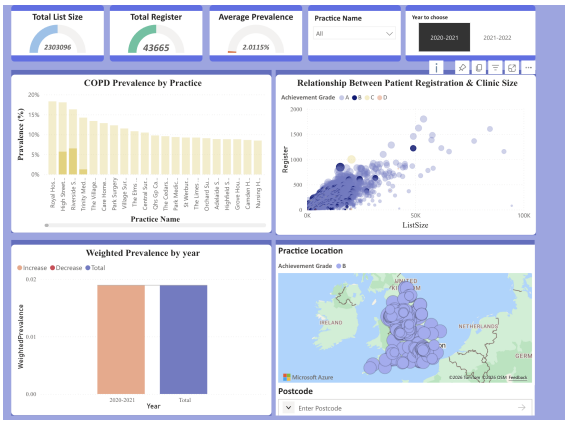
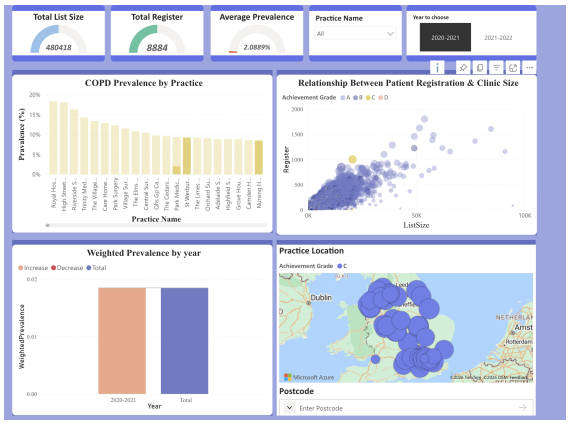


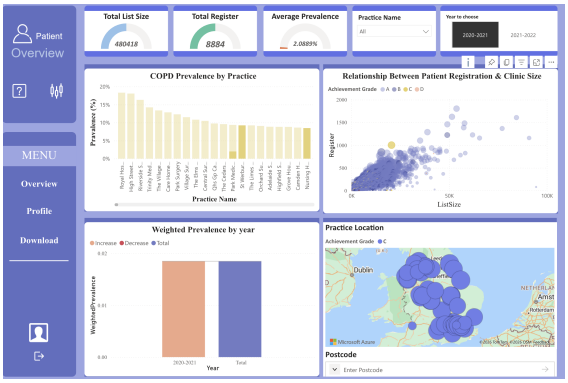
Figure 1: Overview dashboard view



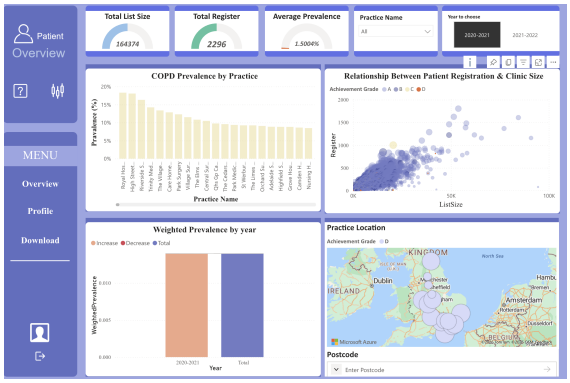
(a) Overview panel A



(b) Overview panel B



(c) Overview panel C



(d) Overview panel D

Figure 2: Overview dashboard components

2.1 Patient-Level View

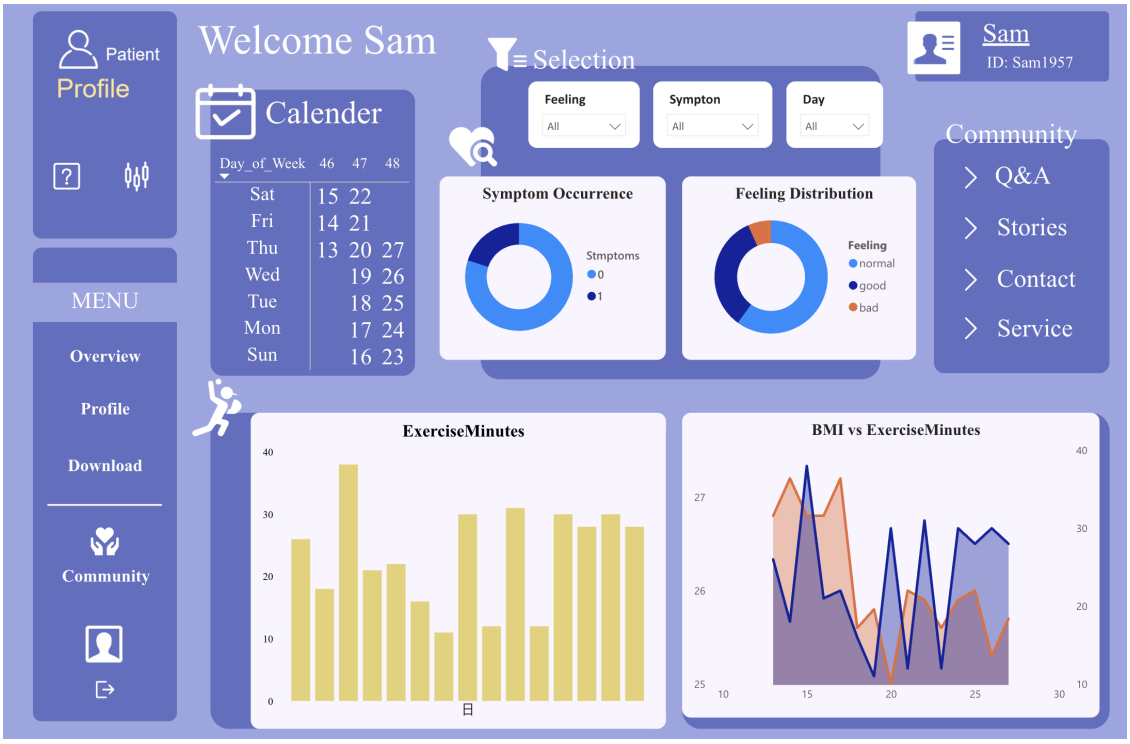


Figure 3: Patient individual-level dashboard view

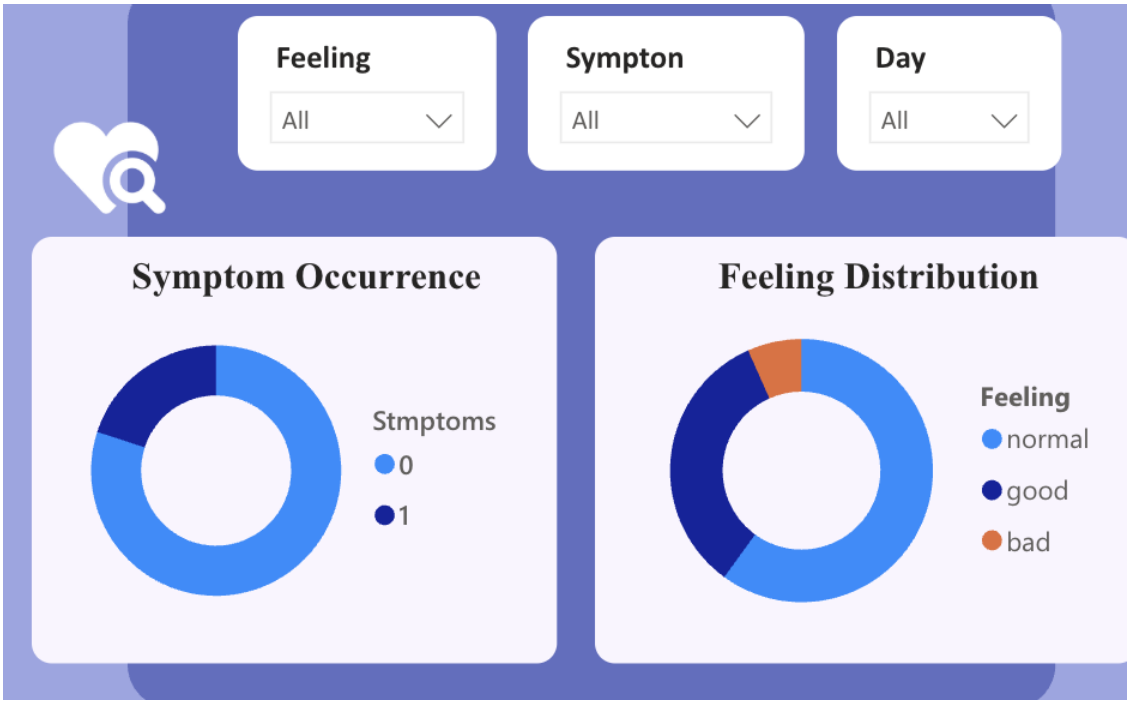


Figure 4: Patient mood dashboard view

As shown in Figure 2, the overview dashboard combines several visual and interactive elements to summarise patient-related data across multiple dimensions. The inclusion of KPI cards, distribution charts, comparative plots, and map-based information suggests that the dashboard was designed to encourage exploratory analysis and provide a comprehensive snapshot of the dataset. This breadth is a clear strength, as it enables users to examine patterns from different perspectives within a single environment. Nevertheless, the simultaneous presentation of numerous visual elements reduces visual hierarchy and may hinder rapid interpretation. Rather than immediately guiding the viewer towards the most relevant insights, the layout requires users to actively navigate and interpret the information for themselves.

3 Initial Impressions

At first glance, the dashboard appears visually rich and feature-complete, offering multiple perspectives on the data. The use of diverse chart types and interactive components suggests an intention to provide a comprehensive analytical tool.

However, the high density of visual elements makes it difficult to immediately identify key insights. The absence of a clear focal point or hierarchy means that users may need additional time to interpret the information presented.

4 Key Findings: Areas for Improvement

4.1 Visual Density

A large number of visual elements are displayed simultaneously. While this increases the amount of information available, it may overwhelm users and reduce clarity, particularly during initial interaction.

4.2 Lack of Visual Hierarchy

There is no clearly defined prioritisation of information. Key metrics and supporting visuals are presented with similar visual weight, making it difficult to determine where to focus attention first.

4.3 Mixed Use of Chart Types

The dashboard employs a wide range of visualisations, including donut charts, scatter plots, and waterfall charts. While each chart type has its purpose, the combination may increase cognitive load and reduce interpretability.

4.4 Navigation and Interaction Complexity

The inclusion of multiple buttons, icons, and navigation elements suggests strong interactivity. However, this may introduce complexity for new users, who may not immediately understand how to navigate or filter the data effectively.

5 Assessment

5.1 Strengths

- Incorporates a broad range of data perspectives
- Provides interactive filtering options
- Demonstrates effort to explore multiple analytical dimensions

5.2 Limitations

- Key insights are not immediately visible
- Users are required to interpret visualisations independently without guidance
- Visual clutter may reduce overall usability

6 Analytical Effectiveness

While the dashboard includes a variety of visualisations, its current structure does not strongly support rapid insight generation. The absence of clear guidance, prioritisation, or highlighted findings means that users must actively search for patterns rather than being guided toward them.

As a result, the dashboard functions more as a collection of visualisations than as a decision-support tool.

7 Challenges

The main challenges identified in this initial assessment include:

- Lack of visual hierarchy
- Overuse of complex or less effective chart types
- Visual clutter from excessive elements
- Limited communication of insights
- Inefficient layout and structure

8 Relevance to Healthcare Context

In healthcare environments, dashboards are often used to support clinical and operational decision-making. The ability to quickly interpret data is critical.

9 Dashboard Improvements

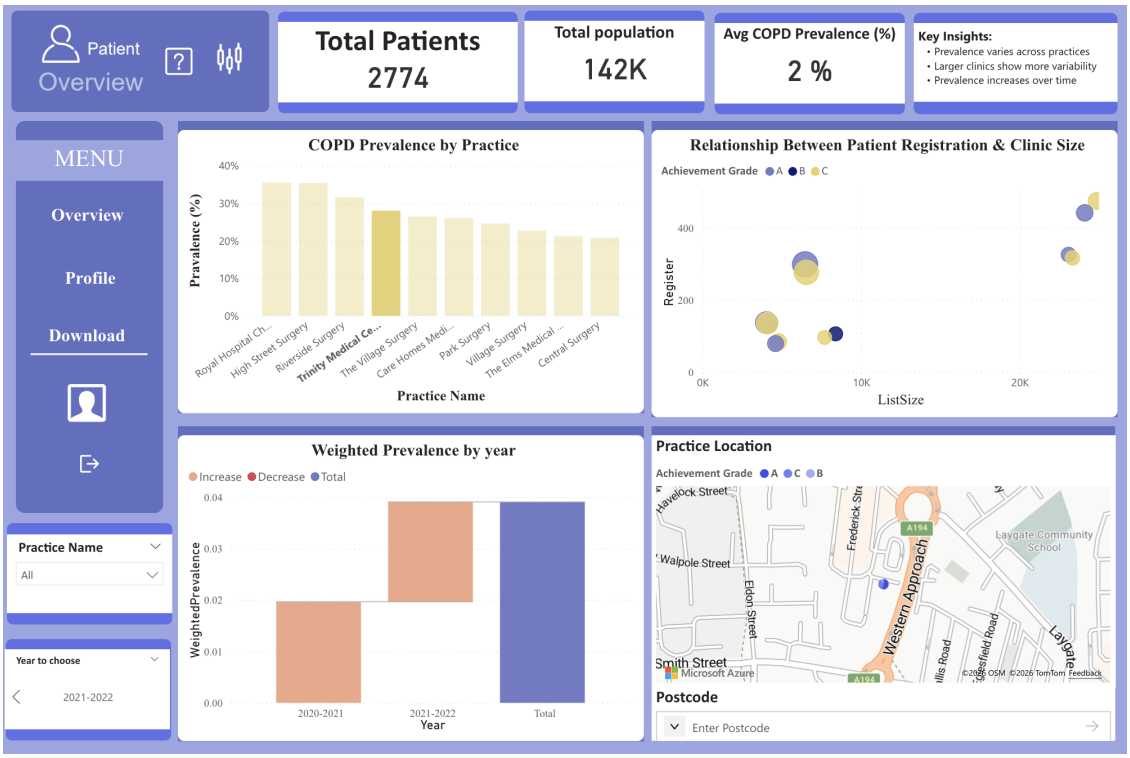


Figure 5: Patient Dashboard Improvement

Based on the initial evaluation, several modifications were implemented to improve the dashboard's usability and analytical effectiveness.

First, gauge visuals were replaced with KPI cards to enhance readability and reduce visual clutter. This allowed key metrics to be interpreted more quickly. A clearer visual hierarchy was introduced by prioritising the most important metric and organising supporting information accordingly.

Second, the number of visual elements was reduced and chart types were simplified. In particular, the bar chart displaying COPD prevalence was refined to focus on the most relevant practices, improving clarity and interpretability.

Third, filters were repositioned to a dedicated section to improve navigation and avoid interference with key information. This change supports a more intuitive user experience.

Finally, a "Key Insights" section was added to guide interpretation and highlight important patterns in the data. This transformed the dashboard from a collection of visualisations into a more effective decision-support tool.

These changes improved readability, reduced cognitive load, and enhanced the dashboard's ability to communicate meaningful insights.